

A PROJECT TITLE

ON

Travel Time Prediction Using Ridge and Lasso Regression
project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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Travel Time Prediction Using Ridge and Lasso Regression

1. ABSTRACT

Accurate travel time prediction is a critical component of intelligent transportation systems, enabling efficient route planning, traffic management, and informed decision-making for commuters. Travel time is influenced by various factors, including travel distance, traffic conditions, time of travel, road characteristics, and other environmental and operational parameters. Due to the complex relationship between these factors, developing reliable predictive models is essential for estimating travel duration accurately. This project presents a machine learning-based approach for Travel Time Prediction Using Ridge and Lasso Regression.

The proposed system employs Ridge Regression and Lasso Regression, which are regularized linear regression techniques designed to improve model stability and generalisation when multiple predictive features are involved. The dataset is initially subjected to preprocessing operations, including handling missing values, removing inconsistencies, feature selection, and appropriate transformation of input variables. The prepared dataset is divided into training and testing subsets, followed by feature scaling to ensure that the variables contribute appropriately to the regression models.

Ridge Regression incorporates L2 regularization to minimise the impact of multicollinearity and prevent excessive coefficient values, whereas Lasso Regression applies L1 regularization, which can reduce the coefficients of less significant features to zero and thereby perform implicit feature selection. Both models are trained and evaluated using standard regression performance measures, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score.

The performance of Ridge and Lasso Regression is comparatively analysed to identify the model that provides superior predictive accuracy and generalisation. The project demonstrates the effectiveness of regularized regression techniques in travel time estimation and highlights their potential application in intelligent transportation and data-driven mobility systems. The developed approach can contribute to improved travel planning, reduced uncertainty in journey duration, and more efficient transportation management.

2. INTRODUCTION

Travel time prediction is an important application of machine learning in intelligent transportation systems. Accurate estimation of travel duration can help commuters plan journeys, select efficient routes, reduce delays, and support effective traffic management. Since travel time is influenced by factors such as distance, traffic conditions, time of travel, and road characteristics, machine learning techniques can be used to identify patterns in historical data and predict travel duration accurately.

This project focuses on applying Ridge Regression and Lasso Regression for travel time prediction. Both techniques use regularization to reduce overfitting and improve model generalisation. Ridge Regression uses L2 regularization, while Lasso Regression uses L1 regularization and can also perform feature selection. The models are trained and evaluated using appropriate performance metrics, and their results are compared to determine their effectiveness in predicting travel time.

3. SOFTWARE REQUIREMENTS

Operating System: Windows / Linux / macOS

Programming Language: Python 3.x

Platform: Jupyter Notebook / Google Colab

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

Dataset: Travel time dataset in CSV format

Hardware: Minimum 4 GB RAM and 5 GB available storage

Signature of the Guide

Course Instructor